

ELYM 328 Laboratory

Experiment 2. Basic Assembly Programming using PIC16F84 microcontroller

Aim: To develop simple assembly programs for PIC16F84 microcontroller and to use MPLAB simulator to display the outputs.

Equipment: Microchip MPLAB IDE software.

Reference materials: MPLAB IDE Quick Start Guide, PIC16F84 Data Sheet and PICmicro Mid-Range MCU Family Reference Manual.

Problem Statements

Problem 1(a). Write an assembly program to light up alternate LEDs at PORTB of PIC16F84. Use MPLAB simulator to check that your program works.

Problem 1(b). The following pattern is desired a set of LEDs connected to the PORTB: ON-ON-OFF-OFF-ON-ON-OFF-ON. Write an assembly programme to achieve this. Use MPLAB simulator to check that your program works.

Problem 2. Write an assembly program to execute the following expression and to display the results at PORTB of PIC16F84:

$$7 + 15 + 208$$

You are required to:

- make use of appropriate mnemonics (assembly language functions) for adding numbers.
- show that your program works. Your test method may include (but not limited to) the use of MPLAB simulator.

Procedure

1. Create a new project for each of the problems above.
2. Add an *assembler* template file to the project for PIC16F84. Give the template file an appropriate name.
3. Also add a *linker* file for PIC16F84.
4. Use the assembly language (PIC16F84.ASM) template file as a starting point for developing your programme.
5. **NB: Please do not modify the original assembly language template file. Rather, save and make use of a copy of the template file for developing your programme.**
6. Configure the port B as an output (PORTB of PIC16F84 is bidirectional, meaning that it can act as input and output. To display results on this port, the port must be configured as outputs using TRISB register).
7. Build and debug your program.
8. Use MPLAB simulator to check the results.

Discussion

1. Explain how you implemented your program for Problem 1(b) above.
2. Explain how your program for the above Problem 2 works. State the purpose of each line of code for the program.
3. Describe the purpose of the following for PIC16F84 microcontroller:
(a) bank memory

- (b) TRISB
- (c) bits 2, 3 and 5 of the STATUS register

Requirements for lab report

9. The report should be written in a standard format. Among other things, your report should include: Title of Experiment, Aims, Introduction, Procedure, Results, Discussion and Conclusion.
10. You are required to attach a copy of your assembly programs for the three problems stated above.
11. Your report should also include screen shots of your output window and the results or outputs that you obtained from MPLAB simulator. Any other relevant figures can be added.

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